

Unit 8, Lesson 2: Using Properties of Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

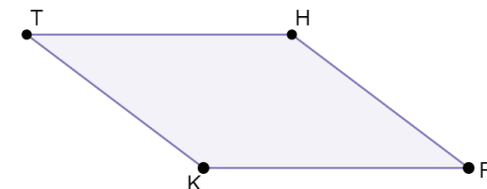
- ☐ I can determine if a quadrilateral is a parallelogram using properties of a parallelogram.



8.2.1 Warm-Up: Bell Work

Given: $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$

Prove: $THRK$ is a parallelogram.



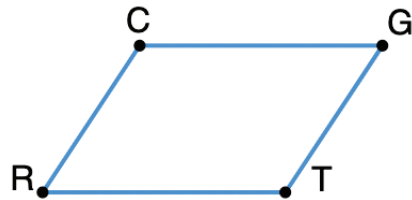
Statement	Reason
1. $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$	1. Given
2. $m\angle RKT = m\angle RHT$ and $m\angle KTH = m\angle KRH$	2. Definition of Congruence
3.	3. The sum of the angles in a quadrilateral is 360° .
4. $m\angle RKT + m\angle RKT + m\angle KTH + m\angle KTH = 360^\circ$ $m\angle RHT + m\angle RHT + m\angle KRH + m\angle KRH = 360^\circ$	4.
5. $2m\angle RKT + 2m\angle KTH = 360^\circ$ $2m\angle RHT + 2m\angle KRH = 360^\circ$	5. Addition
6.	6.
7. $\angle RKT$ is supplementary to $\angle KTH$. $\angle RHT$ is supplementary to $\angle KRH$.	7.
8.	8. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
9. $THRK$ is a parallelogram.	9.



8.2.2 Collaboration: Determining Parallelograms

Work with your partner to determine if the figure given is a parallelogram. Finish the statements that follow.

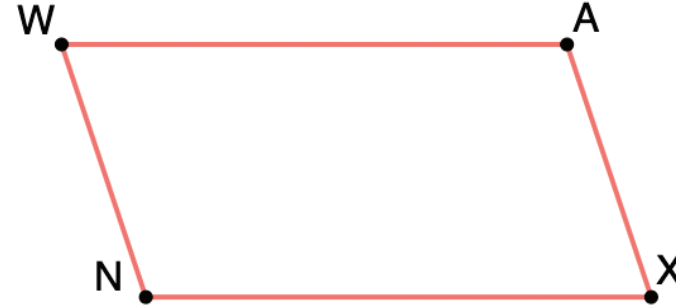
1. $CGTR$ is shown. In the figure, $m\angle C = 113^\circ$, $m\angle G = 67^\circ$ and $m\angle T = 103^\circ$.



a. I can conclude ...

b. I know this because ...

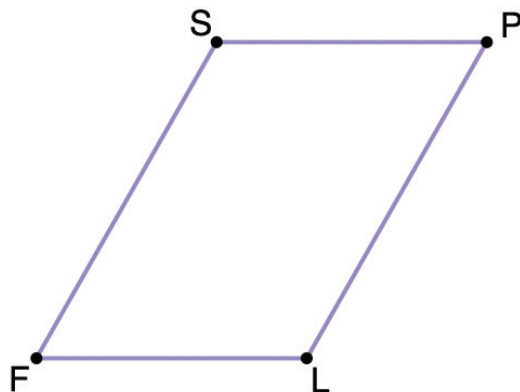
2. $WCXN$ is shown with sides $\overline{WC}$ and $\overline{NX}$ each measuring 15 cm and sides $\overline{WN}$ and $\overline{CX}$ each measuring 9 cm.



a. I can conclude ...

b. I know this because ...

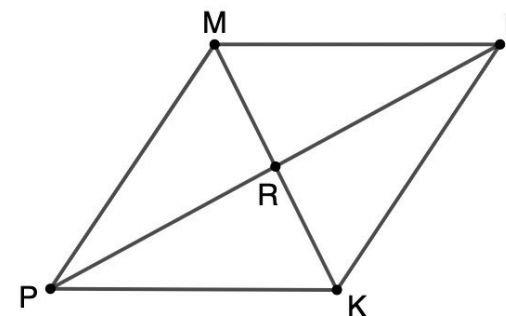
3. In $SPLF$, $m\angle S = 120^\circ$, and $m\angle F = (20x + 47)^\circ$. If $x = \frac{1}{2}$, determine whether the figure is a parallelogram.



a. I can conclude ...

b. I know this because ...

4. $MFKP$ with diagonals $\overline{MK}$ and $\overline{PF}$ intersecting at point R is shown.



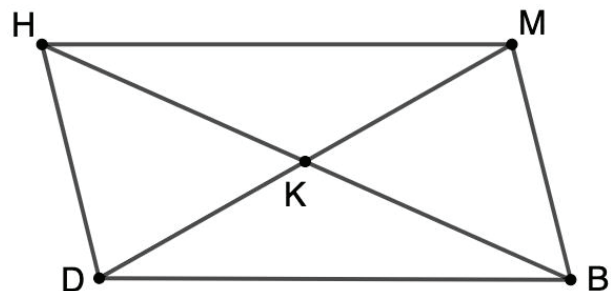
Given that $MR = 4x + 7$, $KR = 3y - 2$, $FR = 4y - 3$, and $PR = 7x - 1$, find the values of x and y that make $MFKP$ a parallelogram.

$x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$



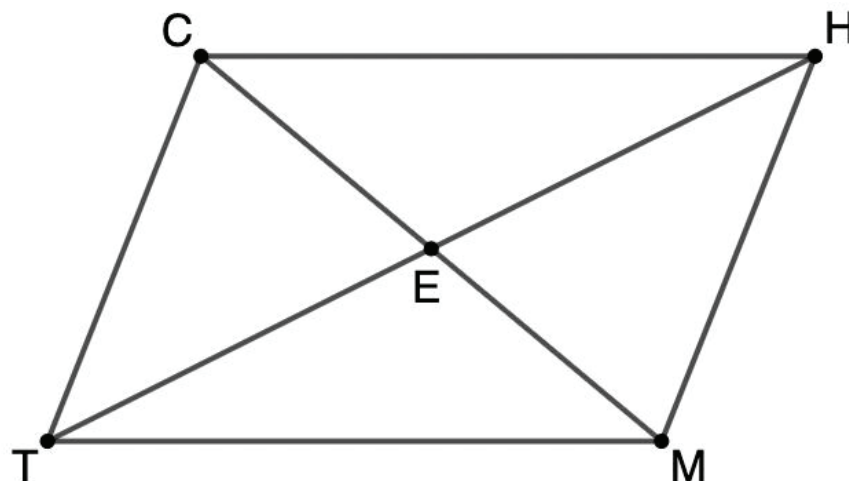
8.2.3 Guided Practice: Is It a Parallelogram?

1. Use the figure given to fill in the chart below.



If you knew ...	...would $HMBD$ be a parallelogram?	Why or why not?
$\angle DHK \cong \angle MBK$ and $\angle BDK \cong \angle HMK$	☐ yes ☐ no	
$\overline{HM} \cong \overline{DB}$	☐ yes ☐ no	
K is the midpoint of $\overline{DM}$ and $\overline{HB}$	☐ yes ☐ no	
$m\angle DHK + m\angle BDK = 180^\circ$	☐ yes ☐ no	

2. Rocio recently inherited a large field in the shape of a quadrilateral from her Uncle Asher. $CHMT$ with diagonals $\overline{CM}$ and $\overline{HT}$ intersecting at point E is shown. She wants to partition the field into four triangular sections to make gardens. While she was out measuring for her supply order, she noticed that the field was shaped like a parallelogram. Her Uncle Asher disagrees. Rocio created the diagram below with the following measurements: $m\angle CHT = 23^\circ$, $m\angle CET = 72^\circ$, $m\angle CMT = 49^\circ$, and $m\angle HTM = 23^\circ$.

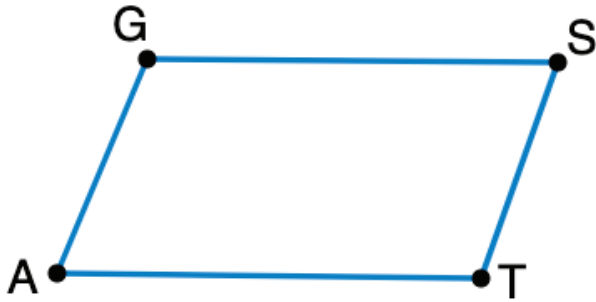


- a. Who is correct?
- b. How do you know?



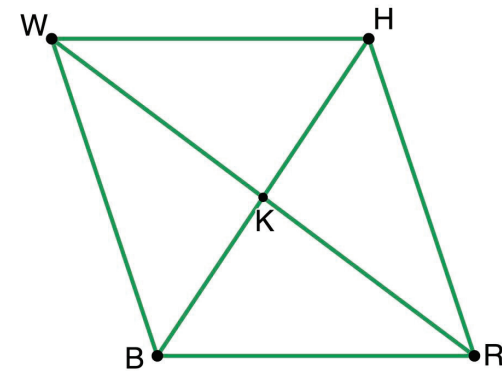
8.2.4 Try It: Parallelograms and Algebra

1. Quadrilateral $GSTA$ is shown where $GS = 3x - 1$, $ST = 2y - 3$, $AT = 11$, and $GA = 8$.



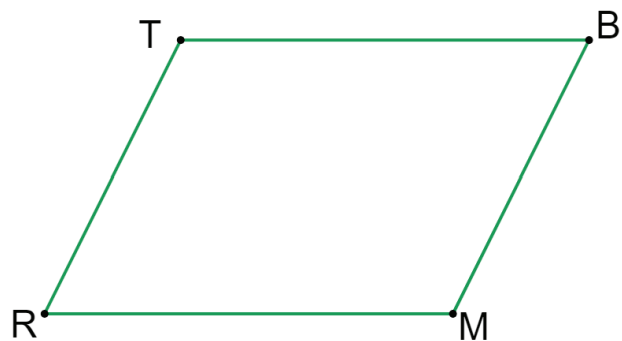
- If $x = 4$ and $y = 6$, determine whether $GSTA$ is a parallelogram.
- If $GSTA$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?

2. $WHRB$ is shown, where diagonals $\overline{WR}$ and $\overline{HB}$ intersect, such that $WK = -y + 4$, $RK = \frac{1}{3}x + 5$, $HK = 5x - 6$, and $BK = 11 + y$.



- If $x = 3$ and $y = -2$, determine whether $WHRB$ is a parallelogram.
- If $WHRB$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?

3. Tariq and Agnes were having a discussion about the different properties of parallelograms. Together they drew and labeled $TBMR$, where $m\angle RTB = (5x - 2)^\circ$, $m\angle TBM = y^\circ$, and $m\angle BMR = 118^\circ$. What are the values of x and y for Tariq and Agnes' figure to be a parallelogram?



8.2.5 Wrap-Up: Cool Down

1. Check the property that is always true about the polygon.

Properties of Figure	Types of Polygons	
	Quadrilateral	Parallelogram
Only one set of parallel sides	☐	☐
Two sets of parallel sides	☐	☐
Two sides of equal length	☐	☐
Consecutive angles are supplementary.	☐	☐
Diagonals are congruent.	☐	☐
The sum of all angles is 360° .	☐	☐
Diagonals bisect each other.	☐	☐
Opposite angles are congruent.	☐	☐